Table 6.2 Comparison of BJT and JFET		
S.No	**BJT**	**JFET**
1.	Bipolar device (current conduction by both type of carriers i.e. majority and minority -electrons and holes)	Unipolar device (current conduction is only due to one type of majority carrier either electron or hole)
2.	Current driven device	Voltage driven device
3.	Low Input impedance	High Input impedance
4.	High noise level	Low noise level
5.	Low Power gain	High Power gain
6.	Low switching speed	High switching speed
7.	Less thermal stability	Better thermal stability
8.	Emitter and collector terminals are not interchangeable	Source and drain terminals are interchangeable

Application

1. The JFET is used as a constant current source.
2. It is used as buffer amplifier.
3. It is used as electronic switch.
4. It is used as phase shift oscillator.
5. JFET is used as a voltage variable resistor (VVR).
6. It is used as high impedance wide band amplifier.

6.10. MOSFET

The MOSFET is an abbreviation of Metal Oxide Semiconductor Field Effect Transistor. In MOSFET, the gate is insulated from the channel by using SiO_2 layer. The input impedance of MOSFET is high, because the gate current is extremely small. It is also called as Insulated Gate FET (IGFET).

History: MOSFET, largely superseded the JFET and has profound effect on digital electronic development, which was invented by Dawankahng and Martin Atalla in 1959. There are two types of MOSFET. They are

1. Enhancement MOSFET (E-MOSFET).
2. Depletion MOSFET (DE-MOSFET).

6.10.1 Enhancement MOSFET (E-MOSFET)

The enhancement MOSFET works only in enhancement mode. It does not conduct when gate to source voltage (V_{GS}) is equal to zero, therefore it is called as 'normally OFF MOSFET'. It is widely used in digital circuits. There are two types of enhancement MOSFET. They are

1. N-channel E-MOSFET
2. P-channel E-MOSFET

6.10.1.1 N-Channel E-MOSFET

Figure 6.11 shows the construction and symbol of n-channel E-MOSFET. It consists of lightly doped *P* type substrate into which two highly doped N-type regions are

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